

# MATTHEW T. ADRIAN

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## Education

### Johns Hopkins University

*Ph.D. Chemical and Biomolecular Engineering (GPA: 3.833/4.0)*

August 2025 - Present

Baltimore, MD

### University of California, Berkeley

*B.S. Chemical Engineering, Concentration: Biotechnology (GPA: 3.588/4.0)*

Degree Conferred: May 2023

Berkeley, CA

## Research and Work Experience

### Johns Hopkins University

*Graduate Researcher, Dr. Jeffrey Gray's Lab*

October 2025 – Present

Baltimore, MD

- Constructed a massive sampling based protein complex prediction pipeline for use in the CAPRI protein-protein docking prediction competition.
- Implementing diffusion energy estimation into AlphaFold3, in progress.

### Merck & Co.

*ML Researcher – Drug Discovery*

August 2023 – August 2025

South San Francisco, CA

- Constructed a denoising method to reduce the noise in drug discovery data, improving deep learning model performance up to 23%.
- Developed a finetuning method improving the predictive performance of chemical pretrained models up to 0.1 Pearson  $r^2$  for ADMET property prediction in both pretrained and non-pretrained baselines.
- Mentored 9 UC Berkeley students to improve on-target potency predictions of graph neural networks using features taken from physics-based docking and ML-based co-folding methods.
- Developed and validated an imputation method to enable a matched molecular pair lead optimization tool to optimize 3+ properties at once.

### Thermo Fisher Scientific

*Chemical Engineering and Operations Intern*

Summer 2021 and 2022

Pleasanton, CA

- Troubleshooted an ultrafiltration unit to remove endotoxin in the downstream oligonucleotide manufacturing process and designed an alternative proposal using chromatography, projecting to save 40+ hours in run time per batch.
- Transferred and optimized 30+ HPLC methods to UHPLC methods using Chromeleon software saving 70%+ in analyte usage, 65%+ in eluent usage, and 55%+ in run time totaling to an estimate of \$45K+ savings per year. Wrote and internally published validation documents and SOPs for the UHPLC instruments to enable accurate and safe usage.

## Publications

( Google Scholar)

1. **M. Adrian**, Y. Chung, & A. C. Cheng. "Denoising Drug Discovery Data for Improved Absorption, Distribution, Metabolism, Excretion, and Toxicity Property Prediction" *J. Chem. Inf. Model.*, **64** (16), 6324-6337 (2024); 10.1021/acs.jcim.4c00639.
2. **M. Adrian**<sup>†</sup>, Y. Chung<sup>†</sup>, K. Boyd, S. Paliwal, S. Veccham, A. C. Cheng. "Multitask finetuning and acceleration of chemical pretrained models for small molecule drug property prediction" *arXiv*, (2025); 10.48550/arXiv.2510.12719. (in prep.)

## Conferences and Invited Talks

1. **M. Adrian**, Y. Chung, & A. C. Cheng. "Denoising Drug Discovery ADMET Data for Improved Regression Task Performance" at ICLR 2024 DMLR Workshop; Vienna, AT; May 2024.
2. C. Suen, B. Lee, **M. Adrian**, et al. "A Multi-Modal Deep Learning Model for Drug Potency Prediction: Leveraging Features from Physics-Based Docking and Advanced Co-Folding Methods" at NeurIPS 2025 AI4Science, FM4LS Workshops; San Diego, CA; Dec. 2025.

## Skills

**Software:** Python (Pytorch, Pandas, scikit-learn, etc.), Bash, GitHub, PBS/Slurm, Jupyter Notebook, Anaconda, Docking Softwares (Glide, Gold, Vina), MATLAB, ASPEN, COMSOL, Simulink, Agile Document Control, Microsoft Office  
**Laboratory:** (U)HPLC, AKTA Flux TFF, Sartorius BioStat C Plus Bioreactor, Thermocycler, NMR, SDS-PAGE

## Certifications

- Deep Learning Specialization (Coursera)